Gauss's law

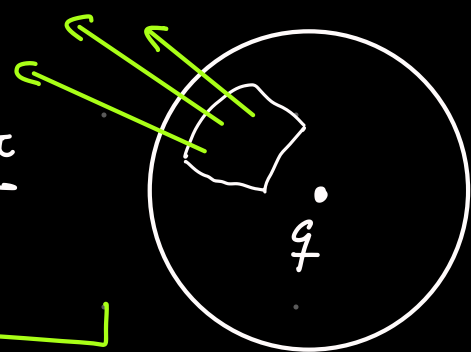
In Maxwell's equation there are two kinds of electric fields;

- The electrostatic field produced by charges
- the Induced electric field produced by Changing Magnetic

Gauss law for electric fields deal with the Electrostatic

Integral Form

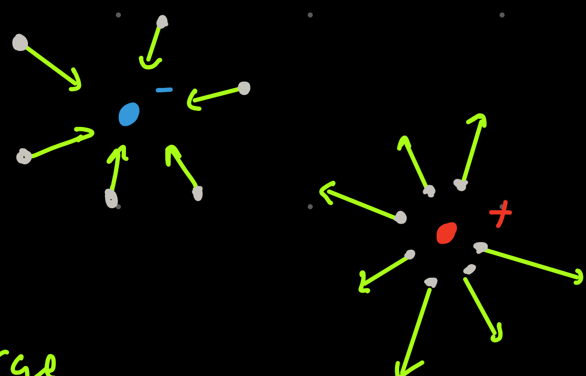
$$\underbrace{\oint_s \vec{E} \cdot \hat{n}\, da}_{\text{Electric flux passing through surface}} = \underbrace{\frac{q_{int}}{\varepsilon_0}}_{\text{amount of charge within said surface}}$$



Differential Form → Deals with points

$$\underbrace{\vec{\nabla} \cdot \vec{E}}_{\text{Divergence of the electric field}} = \frac{\underbrace{\rho}}{\varepsilon_0}$$

Charge density



There is only Divergence
If there is charge

remember the formal Definition of Divergence

$$\vec{\nabla} \cdot \vec{E} \equiv \lim_{\Delta V \to 0} \frac{1}{\Delta V} \oint_S \vec{E} \cdot \hat{n}\, da$$

like $\displaystyle \lim_{\Delta V \to 0} \frac{\phi_E}{\Delta V}$

for a point particle

$$\vec{\nabla} \cdot \vec{E} \equiv \lim_{\Delta V \to 0} \frac{1}{\Delta V} \frac{q}{4\pi \varepsilon_0 r^2} \oint da$$

$$\vec{\nabla} \cdot \vec{E} \equiv \lim_{\Delta V \to 0} \frac{1}{\Delta V} \frac{q}{\varepsilon_0} = \frac{\rho}{\varepsilon_0}$$

Gauss's law for Magnetic fields

Integral Form

$$\underbrace{\oint_S \vec{B} \cdot \hat{n}\, da}_{\text{magnetic flux through a surface}} = 0$$

is always zero

This can be interpreted as there being no Magnetic Monopole, remember that these equations are statements.

Differential Form

$\vec{\nabla} \cdot \vec{B} = 0$ well, this is completely expected